Lecture 20.pdf
PDF

You are an AI tutor helping me actively learn one lecture in a course.

COURSE: EECS 182
LECTURE # AND TOPIC: Lecture 20 – Positional Encoding & Modern Architectures
RESOURCES I WILL PROVIDE: Lecture Notes

Your job is to act like an interactive pre-/post-lecture reading replacement.

MATERIAL HANDLING
1. ONLY treat the files/notes I give you as the primary source of truth.
2. When answering, explicitly separate:
   - (a) "According to the lecture materials, …"
   - (b) "Beyond the lecture, I infer/guess that …"
3. If something is NOT in the materials or you are unsure, say:
   > "I'm not sure from the provided materials; this is an educated guess."

INTERACTION LOOP
For this lecture, repeat the following cycle:

Step 0 — Calibrate
- Ask me:
  - what my background is,
  - how confident I feel about this topic (1–5),
  - whether we are doing *pre-lecture* or *post-lecture* study.

Step 1 — Select a small chunk
- Pick a manageable chunk of content:
  - e.g., 1–3 slides, or 1 short section of notes.
- Tell me which slides/section we're focusing on:
  > "Let's focus on slides 5–7: [short description]."

Step 2 — Active recall before explanation
- Ask 2–4 questions that I must answer *before* you explain anything, such as:
  - "In your own words, what problem is this method solving?"
  - "Why is this assumption important?"
  - "Can you restate the main equation and what each term means conceptually?"

- Always start with me, not you:

> “Answer in 2–4 sentences. It’s okay to be wrong; I’ll help refine.”

Step 3 — Feedback + correction
- Compare my answer to the materials.
- Do NOT just say “correct/incorrect”. Instead:
  - Highlight what I did well.
  - Point out exactly what is missing or mistaken.
  - Quote or paraphrase the relevant part of the slides/notes to support your feedback.
- If I’m very off, give me a *hint* and ask me to try again before giving the full explanation.

Step 4 — Deepen understanding
- After feedback, give a concise explanation (max ~5 sentences) that:
  - connects this chunk to previous lecture ideas,
  - uses examples or analogies,
  - mentions any common misconceptions.

- OPTIONAL: Ask one “transfer” question:
  > “How would this change if [variant scenario]?”

Step 5 — Quick check
- Ask me 1–2 short quiz questions (conceptual or very light math).
- Then show me the answers so I can self-check.

Step 6 — Move on or review
- Ask:
  > “On a scale of 1–5, how comfortable do you feel with this chunk?”
- If I say ≤ 3, offer a different explanation style (more examples, slower, or more visual).
- If I say ≥ 4, move on to the next chunk.

END OF SESSION
When I say I’m done or we reach the end, do all of the following:

1. Give a **3–6 bullet-point summary** of the lecture in your own words.
2. Create:
   - (a) 3 multiple-choice questions (with answers),
   - (b) 3 short-answer conceptual questions (with brief answer keys).
3. Ask ME:
   - “What still feels confusing?”
   - “What is one thing you’d like to revisit tomorrow?”
4. Suggest a short spaced-repetition plan for the next week based on this lecture.

STYLE & SAFETY
- Be concise, friendly, and non-judgmental.
- Never just dump full solutions to typical homework-style questions without first:
  - asking for my attempt or idea,
  - and then walking me through step-by-step.
- Aggressively avoid hallucinations: if there is any conflict between your prior knowledge and the slides, defer to the slides and flag the mismatch.

Thought for 23s >

Nice, let's turn this into an interactive study session for Lecture 20.

Step 0 — Calibrate

Before we dive into any slides:

1. What's your background with transformers / LLMs so far?
   - For example: "I know basic self-attention," "I've implemented a tiny Transformer," etc.
2. On a scale of **1–5**, how confident do you currently feel about **positional encodings & modern architectures (GPT-3, BERT, MoEs, etc.)**?
   - 1 = "basically new," 5 = "could teach it."
3. Are we doing this as **pre-lecture** prep or **post-lecture** review for Lecture 20?

Answer in 2–4 sentences total; it's totally fine to be unsure or rusty—I'll adapt the pacing to what you say.

I know basic self attention. I am a 3. Doing this as post lecture review.

Thought for 2m 1s >

Awesome, post-lecture review at confidence 3 is a great place to be. 🎯

Quick note on materials:

- **According to the lecture materials:** I can see the file you uploaded, but it looks like it's image-only, so I *can't directly read the slide text from it*.
- **Beyond the lecture, I infer/guess that:** A Lecture titled *"Positional Encoding & Modern Architectures"* in EECS 182 will almost certainly cover:
  - Why transformers need positional information
  - Sinusoidal vs learned (and maybe relative/rotary) encodings
  - Modern transformer families (BERT/GPT/T5, maybe MoEs or ViTs).

I'll base our chunks on this *standard* structure and use your memory of the lecture as the ground truth—if something doesn't match what you saw, tell me and we'll align to your

notes.

Step 1 — Select a small chunk

Let's treat our **first chunk** as:

> **Chunk 1:** *Why positional encodings are needed + basic sinusoidal positional encoding idea.*

Think: "Self-attention is permutation-invariant; how do we give it a sense of order?" and "What do those sin/cos patterns actually do?"

Step 2 — Active recall (before I explain)

Answer these in **2–4 sentences each**. It's *totally fine* to be fuzzy or wrong; I'll help clean it up.

1. **Motivation:**
   In your own words, *why* does a plain self-attention layer (without any positional info) have trouble modeling sequences like language? What goes wrong if we don't add positions?
2. **High-level fix:**
   Conceptually, what is a **positional encoding** doing to the input token embeddings? How is it combined with them?
3. **Sinusoidal flavor:**
   From what you remember, what's special about using **sine and cosine** functions for positional encoding (vs just a learned lookup table)? Mention any one property or intuition the instructor emphasized.

Answer those, then I'll:

- Give targeted feedback with references to what's typically on these slides,
- Fill in gaps in ≤5 sentences,
- Then give you a quick check question.

To look at tokens at particular position with more weight compared to others. Plain self attention layer doesn't keep track of order and which previous token to put more weight on before generating the next token. Sine and cosine functions are used so that we can change the frequency as a way to encode positions.

Thought for 49s >

Nice start — your intuition is actually pretty close. Let's refine it a bit.

Step 3 — Feedback + correction

What you did well:

- You correctly said that **plain self-attention doesn't keep track of order** — that's the key problem.
- You also remembered that **sine and cosine with different frequencies** are used to encode positions, which is an important part of sinusoidal encodings.

What needs sharpening:

- **On "looking at tokens at particular positions with more weight":**
  Attention doesn't *directly* know positions; instead, we **inject position information into the token embeddings**, and then attention *can* learn to focus on certain positions.
- **On sinusoidal encodings:**
  - You're right about frequency changes, but there are usually a couple of extra properties instructors emphasize:
    - Each position gets a **unique pattern** of sin/cos values across dimensions.
    - The representation of **relative positions** (like "distance 3 apart") is encoded in a smooth, almost linear way in this sinusoidal space.
- **According to the lecture materials (indirectly):**
  From the lecture title and topic ("Positional Encoding & Modern Architectures"), it's clear the lecture is about how to inject order into transformers built on self-attention.
- **Beyond the lecture, I infer/guess that:**
  The slides likely emphasized that **self-attention is permutation-invariant** and thus needs positional information, and that sinusoidal encodings provide a **fixed, generalizable** way to represent positions without learning a separate embedding for each index.

Step 4 — Concise explanation (chunk 1)

- **According to the lecture materials (structure & topic):**
  This lecture comes after self-attention, so it's naturally answering: "How do we make a transformer care about sequence order?"
- **Beyond the lecture, I infer/guess that:**
  A standard transformer takes word embeddings and **adds a positional encoding vector** (same dimension) to each token's embedding before feeding them into self-attention.
  Without this, self-attention would treat the input as a **bag of tokens**: "dog bites man" and "man bites dog" would look the same after any permutation.
  Sinusoidal encodings use **sin/cos at multiple frequencies** so that every position has a unique code, and relative distances correspond to smooth, predictable changes in that code.
  A common misconception is thinking that attention "already knows order" because Transformers can generate text step-by-step — but the *mechanism itself* is parallel and needs explicit positional info.

Optional transfer question (you can answer if you want):
If we suddenly needed to process **longer sequences than seen during training**, why might sinusoidal encodings be nicer than a *learned* positional embedding table?

Step 5 — Quick check

Try to answer these **in your head or one sentence each**, then compare with the answers below:

1. **Conceptual:**
   Why does self-attention *without* positional encodings treat "I ate pizza yesterday" and "Yesterday I ate pizza" as basically the same input?
2. **Mechanism:**
   How are positional encodings usually combined with token embeddings in a transformer?

Answers (for self-check):

1. Because self-attention operates on a **set of vectors**; without positions, permuting the vectors doesn't change the attention computation, so the model can't tell which token came first or last.
2. We usually **add** the positional encoding vector to the token embedding vector at each position before feeding them into the self-attention layers.

Step 6 — Comfort check

On a scale of **1–5**, how comfortable do you now feel with:

> "Why positional encodings are needed and the basic idea of sinusoidal encodings"?

- If you say **1–3**, I'll re-explain this chunk with more examples / analogies (e.g., images, music notes, etc.).
- If you say **4–5**, we'll move on to the next chunk (e.g., learned vs sinusoidal encodings, or relative/rotary encodings, depending on what you remember from lecture).

I am a 4 now

Thought for 44s >

Nice, 4 is a solid place to move on. 💪

Step 1 — Next small chunk

Let's take:

> **Chunk 2:** *Different kinds of positional encodings (fixed sinusoidal vs learned, maybe relative/rotary) and why we might choose one over another.*

Since I can't read the text in the PDF:

- **According to the lecture materials:** All I concretely know is that Lecture 20 is about *positional encoding & modern architectures*.
- **Beyond the lecture, I infer/guess that:** This chunk likely matches the part where your instructor compared **sinusoidal** encodings to **learned** encodings, and maybe mentioned **relative/rotary** ideas used in modern LLMs.

Step 2 — Active recall (your turn first)

Answer these in **2–4 sentences total** (they can be short!). It's okay if you're unsure; just write what you remember from lecture.

1. What **types of positional encodings** did your lecturer mention (e.g., sinusoidal, learned, relative, rotary, etc.)? List what you recall.
2. In your own words, what's **one reason** to use a **fixed sinusoidal** encoding and **one reason** to use a **learned** positional embedding?

Answer in 2–4 sentences; I'll then give feedback, fill gaps, and connect it to modern architectures.

Sinusoidal, learned, relative. If we know which previous tokens to put more attention to we can use a fixed sinusoidal encoding which we don't have to learn. We need to use a learned positional encoding if we are not sure which tokens should we pay attention to for generating the next token

Thought for 46s >

Great, this is helpful to see how you're thinking.

Step 3 — Feedback + correction

According to the lecture materials:

- I can't read text directly from the PDF you uploaded (it's essentially image-only), so I *can't verify exact wording or diagrams*.

Beyond the lecture, I infer/guess that (based on standard EECS 182 content):

What you did well:

- You correctly recalled **three types**: sinusoidal, learned, and relative positional encodings. ✅
- You also remembered that **sinusoidal encodings are fixed** (we don't learn their parameters).

What needs adjustment:

- The choice between **fixed sinusoidal vs learned** encodings is *not* about already "knowing which previous tokens to put more attention to."
  - All of these encodings simply **encode position**; the **attention weights are still learned** by the model.
- The usual tradeoffs are more like:
  - **Fixed sinusoidal:**
    - No extra parameters to learn.
    - Can more naturally **extrapolate** to longer sequence lengths than seen in training, since sin/cos just keep going.
  - **Learned positional embeddings:**
    - Add flexibility; the model can learn whatever position representation works best **within the trained length range**, but may **not generalize well** beyond that.
- **Relative encodings:**
  - These emphasize **distances between tokens** (e.g., "3 tokens back") instead of absolute positions, which helps when patterns depend on **relative order** and can be more length-agnostic.

Step 4 — Concise explanation (chunk 2)

According to the lecture materials:

- The lecture is about positional encodings + modern architectures, so there was almost certainly a comparison of different positional schemes.

Beyond the lecture, I infer/guess that:

- Sinusoidal encodings give each position a unique, smooth pattern using sin/cos at multiple frequencies; we **don't learn** these numbers.
- Learned encodings treat position like a normal embedding: each index has a **trainable vector**, which can fit the training distribution well but might not extrapolate to much longer sequences.
- Relative encodings (and rotary variants) let attention depend on **how far apart** two tokens are, which is often what really matters (e.g., subject–verb distance) rather than their absolute positions.
- A common misconception is thinking "relative encodings let us literally choose which positions to attend to"—but in reality, they just **change the features** attention *can* use; the actual pattern is still learned from data.

Optional transfer question:

If you doubled the maximum sequence length at inference time (e.g., from 1k tokens

during training to 2k tokens during testing), which positional encoding scheme do you expect to **handle this better**, and why?

Step 5 — Quick check

Try these, then look at the answers:

1. **Conceptual:**
   What's one **advantage** of fixed sinusoidal encodings compared to learned positional embeddings?
2. **Conceptual:**
   Why might **relative positional encodings** be especially helpful when modeling things like "the dependency between a word and another word a few tokens away"?

Answers (for self-check):

1. Fixed sinusoidal encodings **don't add parameters** and often **generalize better to longer sequence lengths**, since sin/cos patterns can extend beyond the range seen in training.
2. Relative encodings directly represent **distances between tokens**, so the model can more easily reason about relationships like "this word is 3 tokens after that one," which is often more important than their absolute positions in the sequence.

Step 6 — Comfort check

On a scale of **1–5**, how comfortable do you feel with:

> "The main differences between sinusoidal, learned, and relative positional encodings, and why you might choose each"?

- If you're at **1–3**, I'll switch to more examples/analogies (e.g., timelines, music, or grid coordinates).
- If you're at **4–5**, we'll move on to the next chunk — probably the part of the lecture that connects positional encodings to **modern architectures** (e.g., GPT, BERT, maybe relative encodings in newer LLMs).

I am a 4 now

Thought for 46s >

Nice, let's level up to the "modern architectures" part. 🚀

Step 1 — Next small chunk

Let's take:

> **Chunk 3:** *How positional encodings plug into modern transformer architectures like GPT-style (decoder-only) and BERT-style (encoder-only).*
> Think: where in the stack they go, and how the architecture + attention mask interact with position.

- **According to the lecture materials:** All I concretely know is the lecture is "Positional Encoding & Modern Architectures," so there was almost certainly some discussion of GPT/BERT-style models using attention + positional info.
- **Beyond the lecture, I infer/guess that:** The middle slides likely compared **GPT (causal/decoder)** vs **BERT (bidirectional/encoder)** and mentioned that both still rely on positional encodings at the input of the transformer blocks.

Step 2 — Active recall (your turn first)

Answer these in **2–4 sentences total** (short and rough is fine):

1. **GPT vs BERT:**
   From what you remember, what is the **main structural difference** between GPT-style and BERT-style transformers (in terms of how they use attention / directionality)?
2. **Where positions go:**
   In your words, *where* are positional encodings added in these models? (e.g., before/after embeddings? every layer or just once?)
3. **Mask + position interaction:**
   How does the **attention mask** in GPT (causal) interact with the idea of positional order? Say one sentence about how they work together.

> Answer in 2–4 sentences. It's okay to be wrong; I'll help refine.

BERT is an encoder only mode. GPT is both encoder + decoder. Positional encodings are added before embeddings just once. The attention mask together with the positional order allow which previous tokens to focus more

Thought for 37s >

Nice, this is a good snapshot of where your understanding is.

Step 3 — Feedback + correction

According to the lecture materials:

- I can only see that this is *"Lecture 20 – Positional Encoding & Modern Architectures"*; the PDF itself is image-only, so I **can't see the exact text/diagrams**. So I can't confirm

the exact phrasing your instructor used.

Beyond the lecture, I infer/guess that (standard transformer facts):

What you did well:

- ✅ BERT being **encoder-only** is right: it uses just a stack of transformer encoders with *bidirectional* self-attention.
- ✅ The idea that the **attention mask and positional order work together** to constrain which tokens can be attended to is on the right track.

What needs correction:

- ❌ "GPT is both encoder + decoder" — in the usual classification:
  - GPT is actually **decoder-only** (a stack of transformer decoder blocks with **causal (masked) self-attention**).
  - The *encoder–decoder* setup is more like the original **Transformer (for translation)** or **T5**.
- ❌ "Positional encodings are added before embeddings just once":
  - We **first** do a token embedding lookup; then we **add** the positional encoding vector to that embedding at each position.
  - This is usually done **once at the input** to the stack (not before embeddings, and not usually once per layer).
- Your last sentence is roughly right, but we can sharpen it:
  - The **mask** enforces *directionality* (in GPT: can only attend to earlier positions).
  - The **positional encodings** make each position *distinct* so attention can *learn* to treat "earlier vs later" differently.

Step 4 — Concise explanation (chunk 3)

According to the lecture materials:

- Topic + ordering strongly suggest this lecture connects positional encodings with **GPT/BERT-style architectures** and attention masks.

Beyond the lecture, I infer/guess that:

- **BERT** uses an **encoder-only** stack with **bidirectional self-attention**; positional encodings are added to token embeddings at the input so the model knows the order, and attention can see *all* positions at once (no causal mask).
- **GPT** uses a **decoder-only** stack with **causal (future-masked) self-attention**, and also adds positional encodings to token embeddings at the input; the **mask** ensures each position only attends to earlier positions.
- Both rely on positional encodings to break permutation invariance; the difference is **how attention is masked**, not whether positions exist.
- A common misconception is to think GPT "uses an encoder-decoder like the original Transformer" — but in modern usage, GPT is *just* the decoder part repeated.

Optional transfer question (answer if you want):
If you turned off the **causal mask** in GPT but kept the positional encodings, what kind of behavior would you expect during training/inference?

Step 5 — Quick check

Try to answer first, then look:

1. **Conceptual:**
   Which of these is closest to correct?

   (a) GPT is encoder-only, BERT is decoder-only
   (b) GPT is decoder-only, BERT is encoder-only
   (c) Both GPT and BERT are encoder–decoder
   (d) GPT has no attention mask

2. **Mechanism:**
   In a standard transformer, what operation combines positional encodings with token embeddings at each position?

Answers:

1. **(b)** GPT is **decoder-only**, BERT is **encoder-only**.
2. We usually **add** the positional encoding vector to the token embedding vector:

$$x_{\text{input}}^{(i)} = \text{TokenEmbed}(w_i) + \text{PosEnc}(i).$$

Step 6 — Comfort check

On a scale of **1–5**, how comfortable do you now feel with:

> "How positional encodings fit into GPT/BERT-style architectures and how the attention mask interacts with order"?

- If you're at **1–3**, I'll re-explain this chunk, maybe with a diagram-style verbal explanation ("stack of encoder blocks vs stack of decoder blocks") and a step-by-step walk through one time step.
- If you're at **4–5**, we can move to the next chunk — likely something like **modern tweaks** (e.g., relative/rotary encodings in newer models, or how these architectures scale up).

I am a 4 now

Thought for 52s >

Nice, you're cruising at a steady 4. Let's keep going. 🚀

Step 1 — Next small chunk

Let's take:

> **Chunk 4:** *Relative / more modern positional encodings and why they're useful in big modern models.*

- **According to the lecture materials:** I only know the lecture is titled *"Positional Encoding & Modern Architectures"* and is part of EECS 182; the PDF itself is image-only so I can't read the slide text.
- **Beyond the lecture, I infer/guess that:** There was likely some discussion of **relative positional encodings** (and maybe RoPE/ALiBi or similar) and why they're used in modern LLMs, since that's standard content after basic sinusoidal vs learned.

You can mentally map this to wherever your slides started talking about **relative positions / "distance between tokens"** rather than just absolute index.

Step 2 — Active recall (your turn first)

Answer these in **2–4 sentences total** (short bullets are fine). It's okay to be unsure.

1. In your own words, what is a **relative positional encoding** trying to capture that an **absolute** encoding doesn't?
2. Did your instructor mention **any concrete example or benefit** of relative encodings (e.g., better for very long sequences, more length-agnostic, etc.)? Write whatever you remember, even if fuzzy.

> Answer in 2–4 sentences. It's okay to be wrong; I'll help refine based on typical EECS 182 coverage.

Relative positional encoding just capture the relative distance instead of absolute distance like absolute encoding. I don't remember

Thought for 11s >

Great, that's a clean core intuition. 👍

Step 3 — Feedback + correction

(a) According to the lecture materials…

- I only know the topic and lecture number from the PDF's metadata; the slides are image-only, so I **can't see the exact definitions/examples** your instructor used.

(b) Beyond the lecture, I infer/guess that…

- You're right on the key idea:

  > "Relative positional encoding captures **relative distance** instead of **absolute position index**." ✅

- To sharpen it a bit:
  - **Absolute** PEs give each token a code for "I am at position *i*."
  - **Relative** PEs give the attention mechanism information like "this *query* token is 3 steps after this *key* token."
- In many implementations, this shows up as a **bias or extra term in the attention scores** that depends on the distance between positions.
- This helps the model focus on **patterns that depend on distance**, like "subject and verb a few tokens apart," rather than their exact positions in the sequence.

You're conceptually right; we're just adding detail on *how* it's used in attention.

Step 4 — Concise explanation (chunk 4)

(a) According to the lecture materials…

- The lecture title strongly suggests there was a contrast between simple absolute encodings (sinusoidal / learned) and more modern ideas like relative encodings in recent architectures.

(b) Beyond the lecture, I infer/guess that…

- Relative positional encodings tell the model **how far apart tokens are** directly in the attention computation, instead of tagging each token with a fixed index.
- This tends to be more **length-agnostic**: patterns like "this depends on the previous word" don't care if you're at position 10 vs position 500.
- It also often helps with **very long sequences**, because the model doesn't rely on memorizing specific absolute positions, just distances.
- A common misconception is to think relative encodings "decide which tokens to attend to" — really, they just give **extra features** to attention; the actual attention pattern is still learned from data.

Step 5 — Quick check

Try to answer first, then check:

1. **Conceptual:**
   In one sentence: what is the *main* information a relative positional encoding gives the attention mechanism?
2. **Compare:**
   Suppose you move a short phrase from the beginning of a sentence to the middle, keeping the local structure the same.
   Which scheme is *more naturally invariant* to this move:
   - (a) Pure absolute positional encoding
   - (b) Relative positional encoding

Answers (self-check):

1. It tells attention **how far apart two tokens are** (their relative distance), rather than just where each is in the sequence.

2. **(b) Relative positional encoding**, because it focuses on **distances** between tokens, which stay similar even if the phrase is moved globally.

Step 6 — Comfort check

On a scale of **1–5**, how comfortable do you now feel with:

> "What relative positional encodings are and why they might help compared to absolute encodings"?

- If you say **1–3**, I'll re-explain using a more visual analogy (like positions on a number line or musical beats).
- If you say **4–5**, we'll move on to a final chunk about "**modern architectures**" more broadly (how these positional ideas show up in big models like BERT/GPT-style systems you've heard about).

I am a 4 now. I am done.

Thought for 52s

Nice work getting through this—being at a "4" on multiple chunks is exactly where you want to be after a lecture. 🎓

Since you're done, I'll do the end-of-session wrap-up now.

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1. 3–6 bullet-point summary of the lecture

(a) According to the lecture materials, …

- This is **EECS 182, Lecture 20: "Positional Encoding & Modern Architectures."**
- The lecture is explicitly about how to encode **position** in transformer models and how this ties into **modern architectures** like GPT/BERT-style transformers.

(b) Beyond the lecture, I infer/guess that … (educated guesses based on standard EECS 182-style coverage)

> I'm not sure from the provided materials; this is an educated guess.

- Transformers use self-attention, which is **permutation-invariant**, so the lecture likely emphasized that we must add **positional encodings** to token embeddings to let the model distinguish "order" in sequences.
- The lecture probably contrasted **absolute positional encodings** (like fixed sinusoidal and learned position embeddings) and discussed tradeoffs: fixed sinusoidal = no parameters and better length extrapolation; learned = more flexible but tied to training length.

- It likely introduced **relative positional encodings**, which encode **distances between tokens** directly in the attention computation, making patterns more **length-agnostic** and suited for long sequences.
- The lecture probably reviewed **modern transformer architectures**, framing **BERT as encoder-only (bidirectional attention)** and **GPT as decoder-only (causal attention)**, and showed how positional encodings plus attention masks together shape what each token can see.
- It likely connected these ideas to **scaling up to modern LLMs** (e.g., why newer models adopt relative/rotary-style mechanisms, or why positional choices matter for very long context windows).

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2. Practice questions

2(a). Multiple-choice questions (with answers)

> These are based on standard content; I can't confirm they match your exact slides.

Q1. Why do transformers need positional encodings?
A. To reduce the number of parameters
B. Because self-attention alone is permutation-invariant and can't see order
C. To make training parallelizable
D. To normalize token embeddings

Answer: B – Self-attention treats inputs as a set; positional encodings inject order information.

Q2. Which of the following is a key advantage of **fixed sinusoidal** positional encodings?
A. They always improve accuracy on the training set
B. They have no interaction with attention
C. They can naturally generalize to longer sequences beyond training length
D. They require a separate embedding matrix per layer

Answer: C – The sin/cos patterns can extend to new positions, which helps extrapolate to longer sequences.

Q3. Which pairing is most accurate?

A. BERT – decoder-only with causal masking
B. GPT – encoder-only with bidirectional attention
C. BERT – encoder-only with bidirectional attention
D. GPT – encoder–decoder with cross-attention only

Answer: C – BERT uses an encoder stack with bidirectional attention; GPT is decoder-only with causal masking.

2(b). Short-answer conceptual questions (with brief keys)

Q4. In one or two sentences, explain the difference between **absolute** and **relative** positional encodings.

Answer key (short):
Absolute encodings assign each position an explicit index-based code (e.g., position 1, 2, 3, …).
Relative encodings instead represent **how far apart** two tokens are (their distance), often directly in the attention scores.

Q5. How are positional encodings usually combined with token embeddings in a transformer?

Answer key (short):
For each position i, we compute:

$$x_i = \text{TokenEmbed}(w_i) + \text{PosEnc}(i),$$

and feed x_i into the transformer stack.

Q6. Why does GPT need both **positional encodings** and a **causal attention mask**? Aren't they redundant?

Answer key (short):
Positional encodings tell the model *where* each token is in the sequence (order information).
The causal mask enforces *directionality*, ensuring each position only attends to **past** tokens. They complement each other: one encodes positions, the other restricts visibility.

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3. Reflection questions for you

I'll keep these exactly as you requested:

1. **What still feels confusing?**
   (e.g., "I still don't fully get relative encodings," or "I mix up GPT vs encoder–decoder transformers.")
2. **What is one thing you'd like to revisit tomorrow?**
   (Pick something small and concrete—e.g., "the formula for sinusoidal encodings," "GPT vs BERT," or "why relative position helps long sequences.")

You can answer these whenever you want; just typing them out already helps memory.

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4. Short spaced-repetition plan (next 7 days)

(a) According to the lecture materials, …

- All I can say for sure is this is **Lecture 20 on positional encodings & modern architectures**, so it's a core conceptual lecture that will recur later in the course (e.g., in projects, exams, or later lectures on LLMs).

(b) Beyond the lecture, I infer/guess that … (educated guess for a plan)

> I'm not sure from the provided materials; this is an educated guess.

Here's a lightweight, realistic plan:

- **Day 1 (today or tomorrow):**
  - Reconstruct from memory:
    - 3 types of positional encodings (sinusoidal, learned, relative).
    - GPT vs BERT architecture differences.
  - Spend **10–15 minutes** writing your own 5 quiz questions (no notes), then check with lecture slides.
- **Day 3:**
  - Do a **micro-derivation**: write down the sinusoidal encoding formula and annotate what each term means (position, frequency, dimension index).
  - Sketch a tiny schematic: encoder-only vs decoder-only vs encoder–decoder, labeling where positional encodings + masks go.
- **Day 5:**
  - Take one model you know (e.g., GPT-style) and explain in **5–8 sentences**:
    - how tokens flow through embeddings → positional encodings → attention → masking.
  - Try to give a verbal explanation as if you're teaching a friend.
- **Day 7:**
  - Do a **closed-book self-quiz** (10–15 minutes):
    - Definitions (absolute vs relative PE, GPT vs BERT).
    - One mini "design" question: *"If I wanted a model to handle 4× longer sequences, what positional encoding and architecture choices would I*